Designer Genes - 2026 Greenbrier Invitational - Division C - Answer Sheet

Team Name: Key **Team #:** Key

Section 1: Multiple Choice (1 point each)

1. C
2. A
3. B
4. B
5. D
6. A
7. D
8. D
9. D
10. A
11. D
12. C
13. D
14. D
15. B
16. B
17. A
18. C
19. B
20. D

Section 2: Short Answer (various points)

21. (1 point each, 4 points total)
    a) BB
    b) Bb
    c) blue horn
    d) white horn

22. GGA-TCA-CTT (3 points)

23. a) incomplete dominance (1 point)
    b) RW x RR (2 points)

	R	W
R	RR	RW
R	RR	RW

(6 points)

Genotypes: RR 50%
RW 50%

Phenotypes: Red flower 50%
Pink flower 50%

(4 points)

24. 40% (1 point)

25. a) yes (1 point)

b)

	A	O
B	AB	BO
O	AO	OO

(12 points)

26. hydrogen (doublet / triple) (1 point)

27. a) AUG-CAA-UUU-CAU (4 points)

b) Met-Gln-Phe-His (4 points)

28. a) UAA, UAG, UGA (3 points)

29. a) I (1 point)

b) Rr (1 point)

c) Smooth Coat (1 point)

d) Ø none (1 point)

30. Centriole (2 points)

31. Messenger ribonucleic acid (3 points)

32. Mother = Rr (1 point)

Father = r^hr (1 point)

	R	r
r^h	Rr^h	r^hr
r	Rr	rr

(6 points)

Genotypes: Rr^h = 25%
Rr = 25%
r^hr = 25%
rr = 25%

Phenotypes: Full color = 50%
himalayan = 25%
albino = 25%

(7 points)

33. a) female (1 point)

b) Trisomy-13 / Patau Syndrome (1 point) ← give credit for either name

c) cleft palate gigantism polydactyly heart abnormalities

(4 points)

34. amino acids (2 points)

35. gap (2 points)

36. a) Non-mendelian (1 point)

b) because it has codominance and multiple alleles.

give credit if at least one reason is mentioned (1 point)

37. a) male dragon = GPdd (1 point) female dragon = GPDd (1 point)

b) can be similified or not.

	Gd	Gd	Pd	Pd
GD	GGDd	GGDd	GPDd	GPDd
Gd	GGdd	GGdd	GPdd	GPdd
PD	GPDd	GPDd	PPDd	PPDd
Pd	GPdd	GPdd	PPdd	PPdd

(12 points)

c) 25% 4/16 (1 point)

d) 13% or 12% (1 point) 2/16

c) 25% (1 point)

38. they should mention polycistronic mRNA, use of operons, or that transcription and translation are coupled (2 points)

39. Promoters (2 points) or promoter region

40. a) introns (1 point)

b) exons (1 point)

c) splicing (1 point)

d) 5'-cap (1 point) or methylated guanine

e) 3'-poly A tail (1 point) or poly adenylated tail, something like.

41. The third nucleotide in a codon can be "wobbly" or different but still code for the same codon. Makes it more efficient (2 points)

42. Is repressed, silenced, turned off, down regulated, etc. (2 points)

43. a) Lactose (1 point)

b) Lactase (1 point)

c) Doesn't waste resources to make (1 point) a protein to digest something that isn't around.

44. the female parent (2 points) also accept mother

45. restriction enzymes (2 points)

46. gel electrophoresis (2 points)

47. a) Retroviruses (1 point) also accept HIV, HTLV or any iteration

b) reverse transcriptase (1 point) also accept RNA-dependent DNA-polymerase

48. Size and Charge (2 points)

↑ also accept Weight